Bobby Hodgkinson

Associate Teaching Professor | Engineering Education | Experiential and AI-Integrated Learning

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Educational Background

M.S., Aerospace Engineering, University of Colorado Boulder, 2011 — Focus: Vehicle Controls
B.S., Aerospace Engineering, University of Colorado Boulder, 2010 — Minor: Electrical Engineering

Academic and Other Relevant Employment History

Associate Teaching Professor *Fall 2024–Present*
University of Colorado Boulder, Smead Aerospace Engineering Sciences
Leads course and curriculum development, experiential laboratory education, and responsible uses of AI for teaching and feedback. Mentors undergraduate researchers, learning assistants, and student developers.

Consultant *June 2026–Present*
Blackboard
Contributes to development of the Scholar Learning Management System, bringing experience in learning-platform design, instructional workflows, and AI-integrated education.

Assistant Teaching Professor *August 2018–Fall 2024*
University of Colorado Boulder, Smead Aerospace Engineering Sciences
Taught and coordinated high-enrollment aerospace laboratory courses; developed experimental hardware, assignments, assessment workflows, and instructional teams across thermodynamics, aerodynamics, structures, dynamics, controls, computing, and design.

Professional Research Assistant *July 2013–August 2018*
University of Colorado Boulder, Smead Aerospace Engineering Sciences
Provided instrumentation and electronics support for teaching laboratories, capstone teams, graduate projects, and research. Designed and validated experimental hardware and software aligned with course learning objectives.

Laboratory Manager *May 2011–May 2013*
University of Florida, Institute for Networked Autonomous Systems
Managed personnel, electronics and machine shops, and improvements to aerial, surface, and underwater robots. Advised undergraduates preparing for the MATE ROV competition.

Honors and Awards

- CU System AI Recognition Award, AI for Teaching & Learning, 2026.
- CU Innovation & Efficiency Award, Top Finalist for *AI-Augmented Personalized Feedback*, 2026.
- Center for Teaching & Learning Excellence in Teaching with Technology Award nomination.
- National Instruments Student Design Showcase, first place, 2010.

Teaching Accomplishments

Classes Taught and Enrollment

Enrollment is summed across all sections for each course and term. The record contains **3,739 student-course enrollments across 24 term-course offerings** from Fall 2020 through current Fall 2026 enrollment. The FCQ workbook documents 78 sections through Spring 2026; Fall 2026 figures are current candidate-reported enrollments.

Term	Course	Course title	Enrollment
Fall 2026	ASEN 3501	Aerospace Experimental Methods (current)	176
Fall 2026	ASEN 1030	Introduction to Computing for Aerospace Engineering (current)	73
Spring 2026	*ASEN 4028*	*Senior Project 2: Design Practicum*	*230*
Fall 2025	ASEN 1030	Introduction to Computing for Aerospace Engineering	51
Fall 2025	*ASEN 4018*	*Senior Project 1: Design Synthesis*	*230*
Summer 2025	ASEN 2803	Dynamics and Controls Laboratory	33
Spring 2025	ASEN 2802	Aerospace Sciences Laboratory I	35
Spring 2025	*ASEN 4028*	*Senior Project 2: Design Practicum*	*221*
Fall 2024	ASEN 2802	Aerospace Sciences Laboratory I	311
Summer 2024	ASEN 2803	Dynamics and Controls Laboratory	28

Term	Course	Course title	Enrollment
Spring 2024	GEEN 1400	Engineering Projects	30
Spring 2024	*ASEN 4028*	*Senior Project 2: Design Practicum*	*214*
Fall 2023	ASEN 2802	Aerospace Sciences Laboratory I	252
Fall 2023	*ASEN 4018*	*Senior Project 1: Design Synthesis*	*214*
Summer 2023	ASEN 2803	Dynamics and Controls Laboratory	22
Spring 2023	ASEN 2802	Aerospace Sciences Laboratory I	20
Spring 2023	ASEN 2803	Dynamics and Controls Laboratory	122
Fall 2022	ASEN 2802	Aerospace Sciences Laboratory I	273
Spring 2022	ASEN 2003	Dynamics and Systems - Lab	264
Spring 2022	ASEN 2012	Experimental/Computational Methods in Aerospace Engineering	31
Fall 2021	ASEN 2002	Thermodynamics and Aerodynamics - Lab	313
Spring 2021	ASEN 2003	Dynamics and Systems - Lab	262
Spring 2021	GEEN 1400	Engineering Projects	27
Fall 2020	ASEN 2002	Thermodynamics and Aerodynamics - Lab	307
		Course enrollments excluding senior-project support	**2,630**
		All enrollments including senior-project support	**3,739**

Italicized rows are ASEN 4018/4028 senior-project support assignments and are excluded from the first subtotal. Fall 2026 values are current enrollments and may change before the end of the term. Earlier teaching not represented in the FCQ workbook includes ASEN 2519 and additional laboratory and project instruction described below.

Selected Course and Curriculum Accomplishments

- **ASEN 3501, Experimental Methods (Fall 2026).** Led extensive development and first delivery of a three-phase, six-outcome course organized around execution and uncertainty quantification, predictive modeling and comparative validation, and experimental design with verification and validation planning. Built a mastery-evidence structure using oral assessment, comparative critique, peer evaluation, and a transparent assessment-to-outcome ontology.
- **ASEN 1030, Introduction to Computing for Aerospace Engineering (Fall 2025).** Designed and taught an inquiry-based course centered on program comprehension, explanation, code interviews, MATLAB Academy credentials, and human-reviewed AI feedback. The shared workflow served approximately 100 students across both sections, processed 969 submissions, returned individualized feedback within 24 hours, and reported median human review time under one minute per response at approximately $7.50 total inference cost for the term.
- **ASEN 2802, Aerospace Sciences Laboratory I (2022–2025).** Led repeated delivery and substantial redevelopment of large-enrollment aerodynamics, structures, and thermodynamics laboratory experiences, including balloon experiments and a combined aeroelasticity activity. Required students to compare analytical, experimental, computational, and AI-generated results and defend discrepancies.
- **ASEN 2002/2003 and ASEN 2803 (2018–2025).** Coordinated multi-section instruction, hardware demonstrations, MATLAB-based analysis, experimental methods, and iterative assessment improvements across thermodynamics, aerodynamics, dynamics, and controls.
- **ASEN 4018/4028, Senior Projects (2013–Present).** Delivered technical workshops and supported project definition, instrumentation, fabrication, software, testing, design review, and evaluation through the Project Advisory Board and formal course roles.
- **ASEN 2012 and GEEN 1400.** Taught Experimental and Computational Methods in Aerospace Engineering and first-year Engineering Projects lecture/laboratory offerings.
- **ASEN 2519, Aerospace CAD/CAM Basics (Maymester 2014 and 2015).** Team-taught SolidWorks, SolidCAM, aerospace machining, manufacturing safety, design practices, and shop techniques through 18 lecture-style sessions in each four-week course for 15–20 students.
- **Additional laboratory instruction (2013–Present).** Delivered recurring introductions and small-group instruction in statics, structures, thermodynamics, aerodynamics, dynamics, vehicle design, and electronics, often for 50–80 students at a time, with additional support across ASEN 2502, 3128, 3200, 3801, 3802, and 5114.

Instructional Materials and Electronic Media

- Created a public shared-source ASEN 3501 course environment embedded in Canvas, including the complete course sequence and a transparent assessment-to-course-learning-outcome ontology.
- Developed an end-to-end Canvas-integrated peer-evaluation system that compiles comments, synthesizes individualized constructive feedback, returns feedback to students, and provides instructors with a summary of notable comments or patterns.
- Developed the human-reviewed ASEN 1030 feedback pipeline and related Canvas retrieval, review, reporting, and

return workflows supporting personalized feedback at scale.

Creator and instructor of a three-course graduate Coursera sequence in generative AI for the CU MS-AI, MS-CS, and MS-DS programs. Learner totals combine public and private course enrollments.

Course	Learners	Link
Introduction to Generative AI (January 12, 2026–Present)	2,714	Course page
Modern Applications of Generative AI (May 4, 2026–Present)	1,020	Course page
Advances in Generative AI (May 4, 2026–Present)	858	Course page
Total learner-course enrollments	**4,592**	

Individual and Group Mentoring

- **Max** (2025), student developer: custom classroom clicker platform; project scoping, implementation planning, and instructional alignment.
- **Saketh** (2025), student developer: knowledge-graph retrieval-augmented-generation chatbot for learning support.
- **Ratna** (2025), student developer: personalized study guides generated from Canvas content extraction.
- **Kaitlyn** (2025), student developer: Canvas-integrated chatbot supporting student learning.
- **Nathan** (CU SPUR/UROP, 2025), undergraduate researcher: reinforcement-prompting concept exploration; a research contribution was incorporated into the ASEN 1030 feedback pipeline.
- **Ronald** (CU SPUR/UROP, 2025), undergraduate researcher: image-recognition concept exploration involving soybean counting and hilum analysis.
- **Charlie Hoke** (2024–2025), teaching-faculty colleague: peer mentoring through departmental work and ACUMent.
- Supervised and trained learning-assistant teams supporting laboratory instruction, course development, equipment validation, safety, and student feedback; mentored aerospace students through Presidents Leadership Class; and mentored two CU Science Discovery high-school students who completed and presented an AI/aeroelasticity investigation.

Leadership and Service Activities

Department

- Chaired teaching-faculty search work in 2022–2023, coordinating application review, interviews, faculty presentation, and recommendation. The 2023 search reviewed more than 30 applications and concluded with a successful hire.
- Sustained educational laboratory infrastructure serving multiple Smead courses, including 24 electronics workstations, software and supplies, vendor relationships, staff transitions, and onboarding needed for instructional continuity.
- Served on working groups for instrumentation, dynamics, thermal/fluid systems, undergraduate curriculum, quality assessment, lab scheduling, and the new curriculum across multiple academic years.
- Faculty advisor for Hyperloop and Sigma Gamma Tau; provided recurring design, project-management, technical-review, academic, and career guidance.
- Served on the Undergraduate Committee, Computing Committee, AeroBEER Engineering Education Research Group, and the senior-capstone Project Advisory Board.

College

- Served on the ITLP Governance Committee and Teaching Professor search work; contributed to College of Engineering and Applied Science generative-AI strategy.
- Served on the dean's ad hoc budget committee, advising on recommendations responding to COVID-19 funding reductions.
- Organized Engineering Social Club events for College of Engineering students and faculty.
- Served on the Capstone COVID Workgroup, developing cross-college strategies and outcome expectations for senior design courses during the pandemic.

University and CU System

- Co-developed separate student and faculty/staff responsible-use courses required for access to ChatGPT EDU, working with approximately six colleagues across CU Boulder and CU Denver and contributing technical guidance, curriculum structure, and AI-literacy content.
- Academic Technology Advisory Group member; contributed to university AI and educational-technology strategy, participated in a small AI/design-based education research working group, and brought Australian AI-literacy practices into CU discussions through professional exchange with Danny Liu.
- Center for Teaching & Learning AI Literacy Ambassador (Spring 2025); delivered and supported campus and college faculty development on responsible generative-AI use, including practical AI-literacy and NotebookLM programming.

- Conversations on Generative AI participant and presenter; engaged in weekly College of Engineering discussions about generative-AI developments and institutional directions.
- Participated in a CTL Students as Partners exploration of student interests and existing CU courses relevant to a possible undergraduate AI certificate.
- Completed the 2025 Online Teaching Academy and applied its online pedagogy and Canvas course-design practices to ASEN 2803.

Community Undertaken on Behalf of the University

- Served on the St. Vrain Valley School District Innovation Center AI Advisory Board and a Boulder Valley School District AI advisory committee, contributing a higher-education perspective on AI literacy and education.
- Designed and delivered practical AI-literacy workshops for Parent Engagement Network and the Louisville YMCA, including *Navigating AI: Build Your Parent Playbook* (November 2025) and an April 2026 guided workshop using ChatGPT, Claude, and NotebookLM.
- Delivered additional AI-in-society and AI-literacy programs for Balfour retirement communities, Boettcher Scholars, Naropa faculty, Women of Aeronautics and Astronautics, Front Range Community College, and other public audiences.
- Make4Covid contributor; fabricated and distributed personal protective equipment during the initial COVID-19 supply shortage.

Professional Organizations

- Rocky Mountain AI Interest Group executive-board member and founder/lead of the AI in Education subgroup; organized recurring educator and community programming and connected regional practitioners, educators, and public audiences around AI literacy.
- Served on the 2025 ASEE Rocky Mountain Section conference planning team and reviewed submissions for ASEE national and regional conferences.

Research and Creative Works

Peer-Reviewed or Paper-Designated Publications and Proceedings

- **Hodgkinson, B.** (2026). "AI-Integrated Inquiry-Based Computing Course." *ASEE Annual Conference*, Paper 50826.
- **Hodgkinson, B.** (2025). "An Aeroelasticity Experiential Learning Activity with AI-Driven Comparative Analysis and Evaluation." *ASEE Annual Conference & Exposition*, Paper 57598.
- Long, C., and **Hodgkinson, B.** (2025). "Evaluating LLM Performance for Generation of Concept Assessment Questions and Individualized Student Feedback in Aerospace Engineering Sophomore Lab." *ASEE Rocky Mountain Section Conference*, Paper 55220.
- **Hodgkinson, B.**, and Whittenburg, N. (2024). "WIP: AI-Based Sentiment Analysis and Grader Enhancements." *ASEE Annual Conference*, Paper 42183.
- **Hodgkinson, B.** (2024). "WIP: Towards an AI Teaching Assistant for Aerospace Engineering Lab Courses." *ASEE Rocky Mountain Section Conference*, Paper 49.
- **Hodgkinson, B.**, and Schwartz, T. (2021). "BYOE: An Apparatus for Exploring Small-Satellite Estimation and Control." *ASEE Annual Conference*, Paper 32904.
- **Hodgkinson, B.**, Lipinski, D., Peng, L., and Mohseni, K. (2014). "High Resolution Atmospheric Sensing Using UAVs." *Distributed Autonomous Robotic Systems*, vol. 104, pp. 31–45. Springer. DOI: 10.1007/978-3-642-55146-8.
- **Hodgkinson, B.**, Lipinski, D., Peng, L., and Mohseni, K. (2013). "Cooperative Control Using Data-Driven Feedback for Mobile Sensors." *IEEE International Conference on Robotics and Automation.*
- Peng, L., Lipinski, D., **Hodgkinson, B.**, and Mohseni, K. (2013). "Dynamic Data Driven Application System for Puff Estimation Using UAVs." *International Conference on Unmanned Aircraft Systems.*
- **Hodgkinson, B.**, Shyu, D., and Mohseni, K. (2012). "Acoustic Source Localization System Using a Linear Arrangement of Receivers for Small Unmanned Underwater Vehicle." *IEEE/MTS OCEANS.*
- **Hodgkinson, B.**, Lipinski, D., Peng, L., and Mohseni, K. (2012). "High Resolution Atmospheric Sensing Using UAV Swarms." *International Symposium on Distributed Autonomous Robotic Systems.*

Other Scholarly Presentations and Publications

The current record does not establish formal peer-review status for the following items.

- Whittenburg, N., and **Hodgkinson, B.** (2025). "Transcription-Enhanced AI Code Review for Deeper Learning." *ASEE Rocky Mountain Section Conference*, Lightning Talk 44.
- **Hodgkinson, B.**, Lipinski, D., Peng, L., and Mohseni, K. (2012). "Environmental Monitoring with K-Means

Error Reduction Using UAVs Controlled by a Fluid-Based Scheme." *IEEE/RSJ IROS Workshop on Robotics for Environmental Monitoring*.

- Krieg, M., Klein, P., **Hodgkinson, R.**, and Mohseni, K. "A Hybrid Class Underwater Vehicle: Bioinspired Propulsion, Embedded System, and Acoustic Communication and Localization System." *Marine Technology Society*, pp. 153–164.
- Ambrosio, D., Gizzi, R. D., **Hodgkinson, B.**, Kirkpatrick, J., Miller, C., Price, J., and Thomas, T. (2011). "CUBOAT: Colorado Underwater Buoyant Oceanic Acoustic Network." *49th AIAA Aerospace Sciences Meeting*.

Invited Talks and Other Conference Presentations

- Kiewit Faculty Open House, invited national presentation translating AI-in-education deployments for approximately three dozen civil-engineering and construction-management faculty, June 8–10, 2026.
- MathWorks Educator Conference Debrief, Boulder, Colorado, January 28, 2026.
- MathWorks Educator Conference, Carleton College, October 25, 2025.
- RMACC AI Workshop, August 5, 2025.
- AI @ CU Boulder Showcase, Boulder, Colorado, April 18, 2025.
- NAIRR Pilot Program Talk, April 2, 2025.
- 2025 ASEE Annual Conference & Exposition.
- 2025 ASEE Rocky Mountain Section Conference, University of Colorado Boulder, May 14–16, 2025.
- ASEE Annual Conference, Portland, Oregon, June 23–26, 2024.
- ASEE Rocky Mountain Section Conference, Boulder, Colorado, May 15–18, 2024.
- CU–KAIST educational collaboration, Daejeon, South Korea; represented CU and presented/shared aerospace spin-module experiential-learning work, May 14–18, 2023.
- ASEE Annual Conference, virtual meeting, July 26–29, 2021.
- IEEE/MTS OCEANS, Hampton Roads, Virginia, October 14–19, 2012.
- AIAA International Student Conference, Orlando, Florida, January 4–7, 2011.
- National Instruments Student Design Showcase, Austin, Texas, August 3–5, 2010 (first place).

Research and Project Support

Year	Source	Project/support	Role and status
2025	NAIRR / Texas Advanced Computing Center	Computational allocation for development and evaluation of a knowledge-graph retrieval-augmented-generation educational chatbot platform.	Principal investigator.
2025	Dean's Excellence Fund	Fellowship-style support for SPUR undergraduate research through the BUFFALOES AI-literacy initiative.	Recipient/project lead.
2025	CU Boulder Embark	Classroom knowledge-graph chatbot and AI-support suite.	Advanced to round two; developmental activity.
2025	MISSE proposal	Education component for a proposal led by PI Tim Minton.	Invited educational director; proposal not funded.

Online Education, Media, and Public Engagement

- Guest, *Smarter Campus Podcast*: "Engineering Understanding: AI, Aerospace, and Rethinking Learning," July 21, 2025.
- Guest, *Parenting Well Podcast*: "Raising Thinkers in the Age of AI," January 7, 2026.
- Guest, *Conscious Millionaire Podcast*: "Personalized Solutions for Science and Tech Customers."
- Featured by CU Boulder in "Adding AI to the Aerospace Classroom."
- Featured in *Colorado AI News*, "Pic of the Week: CU Boulder professor Bobby Hodgkinson brings AI awareness to the senior class," March 22, 2025.